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Coleridge Faraday

Education

PhD in Physics, *University of Cape Town* 2025–Present
- Supervisor: Assoc. Prof. W. A. Horowitz

MSc with Distinction in Physics, *University of Cape Town* (90%) 2023–2024
- Thesis: *Energy Loss and Theoretical Uncertainties in Small Quark-Gluon Plasmas*
- Supervisor: Assoc. Prof. W. A. Horowitz

Honours in Physics, *University of Cape Town* (85%) 2022
- Supervisor: *Investigating the short path length corrections to jet phenomenology in heavy ion collisions*
- Supervisor: Assoc. Prof. W. A. Horowitz

BSc in Physics and Mathematics, *University of Cape Town* (81%) 2019–2021
- Supervisor: *Spontaneous symmetry breaking and the Higgs mechanism.*
- Supervisor: Assoc. Prof. Heribert Weigert

Awards

Best PhD Oral Presentation in Theoretical and Computational Physics at SAIP2025 Annual Conference 2025
National Institute for Theoretical and Computational Sciences (NITHECS) Scholarship 2025 – 2027
Science Faculty PhD Fellowship, University of Cape Town 2025 – 2027
Manuel & Luby Washkansky Scholarship, University of Cape Town 2023
SA-CERN Excellence Scholarship 2023 – 2024
Science Faculty Scholarship, University of Cape Town 2020
Science Faculty Entrance Scholarship, University of Cape Town 2019
Dean's Merit List, University of Cape Town 2019, 2021, *
UCT Mathematics Olympiad Gold Award 2015, 2016
Professor John Webb Mathematics Scholarship, Wynberg Boys' High School 2014 – 2018

Publications

Journal Articles

[1] **Faraday, C.,** Horowitz, W. A. & Schenke, B. **Sept. 2026.** Centrality-dependent nuclear modification from hard-soft correlations in the glasma. arXiv: 2609.29886 [hep-ph]. **2026**

[2] Bert, B., **Faraday, C.** & Horowitz, W. A. **Mar. 2026.** Energy loss predicts no v_2 in small systems. arXiv: 2603.21861 [hep-ph].

[3] **Faraday, C.,** Bert, B., Brand, J., Vogelsang, W. & Horowitz, W. A. **Dec. 2025.** From Lead to Helium: Potential for a Measurement of Jet Quenching in the Smallest Collision Systems. Accepted for publication in Phys. Rev. Lett. doi:10.1103/ykmp-fy57. arXiv: 2512.17832 [hep-ph]. **2025**

[4] **Faraday, C.** & Horowitz, W. A. **Nov. 2025.** Statistical analysis of pQCD energy loss across system size, flavor, $\sqrt{s_{NN}}$, and p_T. *JHEP* **11,** 019. doi:10.1007/JHEP11(2025)019. arXiv: 2505.14568 [hep-ph].

*No awards were given in 2020 due to the COVID-19 pandemic

[5] **Faraday, C.** & Horowitz, W. A. **May 2025.** Collisional and radiative energy loss in small systems. *Phys. Rev. C* **111,** 054911. doi:10.1103/PhysRevC.111.054911. arXiv: 2408.14426 [nucl-th].

[6] **Faraday, C.** & Horowitz, W. A. **Mar. 2025.** A unified description of small, peripheral, and large system suppression data from pQCD. *Phys. Lett. B* **864,** 139437. doi:10.1016/j.physletb.2025.139437. arXiv: 2411.09647 [hep-ph].

[7] **Faraday, C.,** Grindrod, A. & Horowitz, W. A. **Nov. 2023.** Inconsistencies in, and short pathlength correction to, $R_{AA}(p_T)$ in $\mathrm{A}+\mathrm{A}$ and $\mathrm{p}+\mathrm{A}$ collisions. *Eur. Phys. J. C* **83,** 1060. doi:10.1140/epjc/s10052-023-12234-y. arXiv: 2305.13182 [hep-ph]. **2023**

Conference Papers

[1] **Faraday, C.,** Bert, B., Brand, J., Vogelsang, W. & Horowitz, W. A. **Apr. 2026.** *Jet Quenching in the Smallest Hadronic Collision Systems* in *60th Rencontres de Moriond on QCD and High Energy Interactions: Moriond QCD 2026* (Apr.). arXiv: 2604.18010 [hep-ph]. **2026**

[2] Bert, B., **Faraday, C.** & Horowitz, W. A. **Dec. 2025.** *Parameterizing the Geometry of the QGP on an Event-by-Event Basis* in *68th Annual Conference of the South African Institute of Physics* Contribution to SAIP 2025 (Dec.), 998–1003. ISBN: 978-1-0492-1907-3. **2025**

[3] **Faraday, C.** & Horowitz, W. A. **Dec. 2025.** *Energy Loss as a Probe of Quark-Gluon Plasma Formation Across Collision System Size* in *68th Annual Conference of the South African Institute of Physics* Contribution to SAIP 2025 (Dec.), 998–1003. ISBN: 978-1-0492-1907-3.

[4] **Faraday, C.** & Horowitz, W. A. **Jan. 2025.** *Heavy Flavour Energy Loss in Small and Large Systems* in. **316.** Contribution to SQM2024 (Jan.), 04001. doi:10.1051/epjconf/202531604001. arXiv: 2409.14886 [nucl-th].

[5] Bert, B., **Faraday, C.** & Horowitz, W. A. **Dec. 2024.** *Modeling the Geometry of the Quark Gluon Plasma* in *68th Annual Conference of the South African Institute of Physics* Contribution to SAIP 2024 (Dec.), 618–623. ISBN: 978-1-0370-2645-4. **2024**

[6] **Faraday, C.** & Horowitz, W. A. **Dec. 2024.** *Energy Loss in Small Quark Gluon Plasmas* in *68th Annual Conference of the South African Institute of Physics* Contribution to SAIP 2024 (Dec.), 629–634. ISBN: 978-1-0370-2645-4.

[7] **Faraday, C.** & Horowitz, W. A. **June 2024.** *High-pT suppression in small systems* in. **296.** Contribution to Quark Matter 2023 (June), 12011. doi:10.1051/epjconf/202429612011. arXiv: 2312.10712 [hep-ph].

[8] **Faraday, C.** & Horowitz, W. **Feb. 2024.** *Energy Loss in Small Collision Systems* in. **HardProbes2023.** Contribution to Hard Probes 2023 (Feb.), 131. doi:10.22323/1.438.0131. arXiv: 2307.08355 [nucl-th].

[9] **Faraday, C.** & Horowitz, W. A. **Sept. 2023.** *Assumption Breakdown in Radiative Energy Loss* in *67th Annual Conference of the South African Institute of Physics* Contribution to SAIP 2023 (Sept.). arXiv: 2309.06246 [hep-ph]. **2023**

Theses

[1] **Faraday, C. Dec. 2024.** *Energy loss and theoretical uncertainties in small quark-gluon plasmas* MSc thesis (Dec.). arXiv: 2506.02056 [hep-ph]. **2024**

Talks and Posters

Slides for my major presentations are available here under a CC-BY license.

Seminars

[1] **Faraday, C. Oct. 2026.** *From Lead to Helium: Probing Matter at Extreme Temperatures and Sizes* UCT Physics Colloquium. University of Cape Town, Cape Town, South Africa, Oct. **2026**

[2] **Faraday, C. Aug. 2026.** *From Lead to Helium: Searching for the Smallest Quark-Gluon Plasma* Invited pair of seminars at the UCT Centre for Theoretical and Mathematical Physics. University of Cape Town, Cape Town, South Africa, Aug.

[3] **Faraday, C. Aug. 2026.** *Introduction to jet quenching phenomenology in the quark-gluon plasma* Invited pair of seminars at the UCT Centre for Theoretical and Mathematical Physics. University of Cape Town, Cape Town, South Africa, Aug.

[4] **Faraday, C. June 2026.** *Discovery Potential for Jet Quenching in the Smallest Collision Systems* McGill University, Montreal, Canada, June.

[5] **Faraday, C. June 2026.** *Uncovering Jet Quenching in the Smallest Hadronic Collision Systems* University of Illinois Urbana-Champaign, USA, June.

[6] **Faraday, C. Apr. 2026.** *Discovering the smallest droplet of quark-gluon plasma through jet quenching* Rising Researchers Seminar Series, hosted by the University of Washington. Virtual, Apr. https://sites.uw.edu/risingresearchers/seminars/current-seminars/.

[7] **Faraday, C. June 2025.** *Is there energy loss in small systems?* Set of three seminars. New Mexico State University Nuclear Theory Group (Virtual), June. **2025**

Conference Talks

[1] **Faraday, C. Sept. 2026.** *From Lead to Helium: Partonic Energy Loss in Very Light Ion Collisions* EMMI Workshop: Light-ion collisions 2026. Heidelberg, Germany, Sept. **2026**

[2] **Faraday, C. July 2026.** *Jet quenching in the smallest quark-gluon plasma droplet* 70th Annual Conference of the South African Institute of Physics (SAIP 2026). University of the Western Cape, Cape Town, South Africa, July.

[3] **Faraday, C. June 2026.** *First exploration of hard-soft correlations from IP-Glasma* Hard Probes 2026. Vanderbilt University, Nashville, Tennessee, USA, June.

[4] **Faraday, C. Mar. 2026.** *Jet Quenching in the Smallest Hadronic Collision Systems* 60th Rencontres de Moriond 2026: QCD and High Energy Interactions. La Thuile, Italy, Mar.

[5] **Faraday, C. Dec. 2025.** *Light-Ion Suppression from pQCD Energy Loss with Small-System Corrections* Light ion collisions at the LHC - 2025. CERN, Geneva, Switzerland, Dec. **2025**

[6] **Faraday, C. Sept. 2025.** *Statistical Analysis of Energy Loss Across System Size* Precision Frontier of QCD Matter: Inference and Uncertainty Quantification. Central China Center for Nuclear Theory, Central China Normal University, Wuhan, China, Sept.

[7] **Faraday, C. July 2025.** *Energy Loss as a Probe of Quark-Gluon Plasma Formation Across Collision System Size* 69th Annual Conference of the South African Institute of Physics (SAIP 2025). University of the Witwatersrand, Johannesburg, South Africa, July.

[8] **Faraday, C. July 2024.** *Energy Loss in Small Quark Gluon Plasmas* 68th Annual Conference of the South African Institute of Physics (SAIP 2024). Makhanda, South Africa, July. **2024**

[9] **Faraday, C. June 2024.** *Heavy Flavour Energy Loss in Small and Large Systems* The 21st International Conference on Strangeness in Quark Matter (SQM 2024). Strasbourg, France, June.

[10] **Faraday, C. Sept. 2023.** *High-p_T suppression in small systems* 30th International Conference on Ultrarelativstic Nucleus-Nucleus Collisions (Quark Matter 2023). Houston, Texas, USA, Sept. **2023**

[11] **Faraday, C. July 2023.** *Short path length corrections to pQCD phenomenological energy loss models in a QGP* 67th Annual Conference of the South African Institute of Physics (SAIP 2023). Richard's Bay, South Africa, July.

Conference Posters

[1] **Faraday, C. June 2026.** *From Lead to Helium: Discovery Potential for Jet Quenching in the Smallest Collision Systems* Hard Probes 2026. Vanderbilt University, Nashville, Tennessee, USA, June. **2026**

[2] **Faraday, C. Apr. 2025.** *Data-Driven Predictions for Small System Energy Loss* Quark Matter 2025. Goethe University Frankfurt: Quark Matter 2025, Apr. **2025**

Outreach Talks

[1] **Faraday, C. Oct. 2026.** *Matter at the Extreme: the Quark-Gluon Plasma* UCT Physics Postgraduate Showcase 2026. University of Cape Town, Cape Town, South Africa: University of Cape Town, Oct. **2026**

[2] **Faraday, C. Apr. 2026.** *The Hottest Substance in the Universe* NITheCS Bursary Holders Workshop 2026. University of Stellenbosch, South Africa, Apr.

[3] **Faraday, C. Sept. 2025.** *What Happens When Protons Melt?* Cape Town, South Africa, Sept. **2025**

[4] Brand, J., **Faraday, C.** & Roosenstein, R. **July 2025.** *Outreach talk to disadvantaged primary school students* In collaboration with Bartosch Foundation. Masiphumelele, South Africa, July.

[5] **Faraday, C. May 2025.** *Smashing Atoms to Rewind the Universe* The University of Cape Town's Society of Physics Students. Cape Town, South Africa, May.

[6] **Faraday, C. Oct. 2024.** *Racing Balls* Phenomenal physics (high school outreach). Cape Town, South Africa: University of Cape Town, Oct. **2024**

[7] **Faraday, C. Sept. 2024.** *Introduction to Heavy-Ion Physics* Outreach trip to University of Venda. Thohoyandou, South Africa: University of Venda, Sept.

[8] **Faraday, C. Nov. 2023.** *Quark Gluon Plasma Formation in Small Systems* ATLAS Showcase, UCT-CERN relaunch. Cape Town, South Africa, Nov. **2023**

[9] **Faraday, C. Oct. 2023.** *Probing the Early Universe with Extreme Temperature Physics* Postgraduate Showcase 2023. Cape Town, South Africa: University of Cape Town, Oct.

Academic Service

Postgraduate representative for Physics Department, University of Cape Town — 2025 – Present
Organizer, UCT Physics Postgraduate Showcase — 2025, 2026

Reviewing

SCIENCE CHINA Physics, Mechanics & Astronomy
Annals of Physics
Proceedings of the 68th Annual Conference of the South African Institute of Physics
Proceedings of the 69th Annual Conference of the South African Institute of Physics
Proceedings of the 70th Annual Conference of the South African Institute of Physics
ESKOM Science Expo 2025 Project Judge

Teaching

Third Year Particle Physics (UCT), Tutor 2026
Quantum Field Theory II (SATACS), Tutor 2025
Quantum Field Theory I (SATACS), Tutor 2024,2025
Honours Statistical Physics (UCT), Tutor 2024
Honours Quantum Mechanics (UCT), Tutor 2023, 2024
Third Year Statistical Physics (UCT), Tutor 2023,2025
First Year Physics (with laboratory) (UCT), Tutor 2022
High School Physics, Mathematics, Chemistry, and Computer Science (Chit-Chat Tutoring), Private Tutor 2019–2022

Mentorship roles

Ben Bert. Mentoring and helping with honours and MSc projects on phenomenological energy loss at UCT 2023 – Present
Tiaan van der Merwe. Mentoring and helping with honours and MSc projects on phenomenological energy loss at UCT 2024 – Present
Jack Brand. Mentoring and helping with honours and MSc projects on phenomenological energy loss at UCT 2025 – Present
Daniel Broomhall. Mentoring and helping with honours project on phenomenological energy loss at UCT 2026